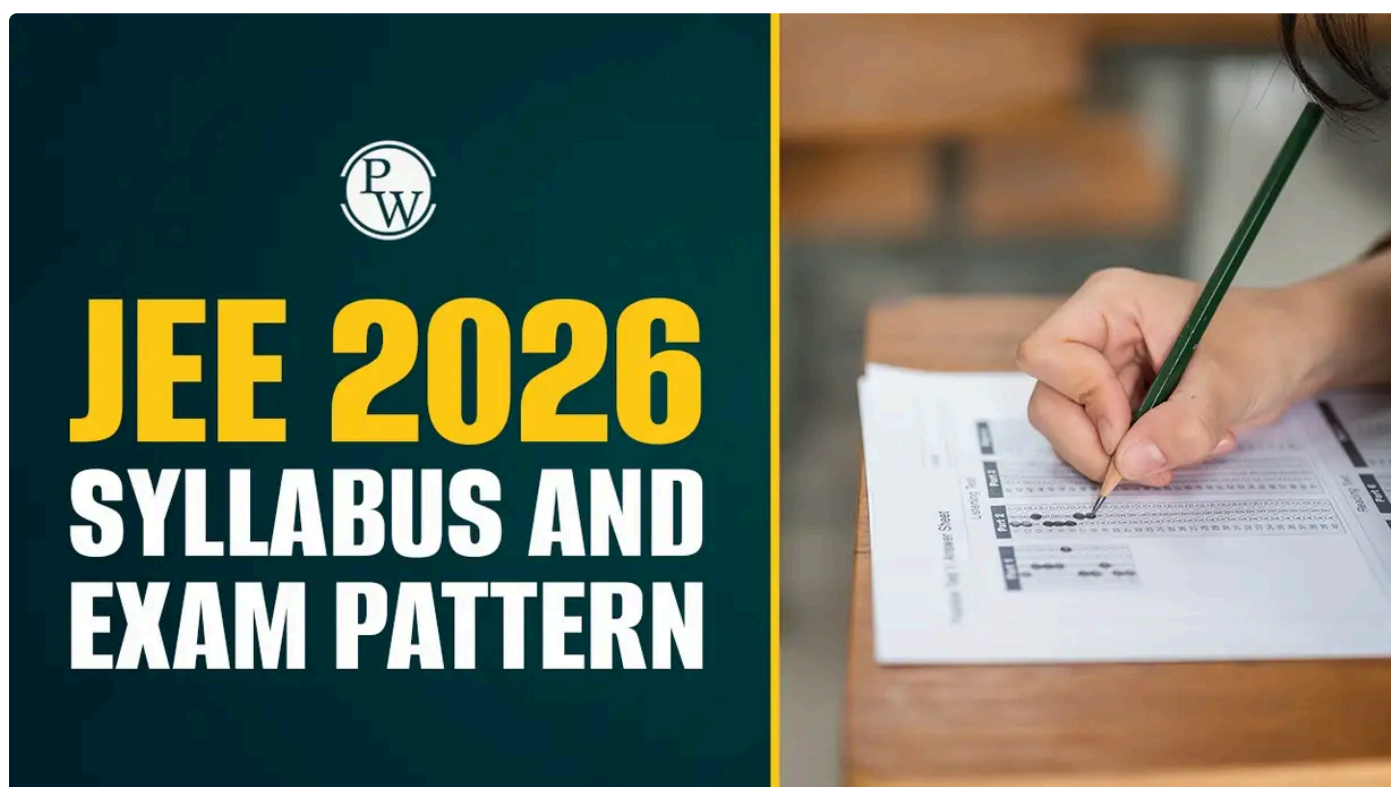




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JEE Syllabus 2026 And Exam Pattern

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JEE Syllabus 2026 covers all the important topics for the Mains and Advanced exam stages to help you prepare better for the upcoming Joint Entrance Examination (JEE). Lakhs of aspirants appear for this exam aiming to get into top engineering colleges like IITs, NITs, and IIITs. So, to stand out from the crowd, you will need a robust study plan that deeply covers the JEE Syllabus 2026.

JEE Syllabus 2026 is based on three main subjects: Physics, Chemistry, and Mathematics. The JEE Main syllabus mainly covers important topics from Class 11 and 12 [IIT JEE books](#), while the JEE Advanced syllabus is a bit more detailed and challenging. So, when you know which topics to focus on and which ones to give less priority from the vast JEE syllabus 2026, it helps you save time and make your preparation more exam-focused.

Along with the IIT JEE syllabus, understanding the JEE exam pattern is also very important. It tells you about the marking scheme, along with the types and number of questions asked in both JEE

Main and JEE Advanced. So, keep reading to know the complete JEE exam syllabus and the detailed exam pattern for both Mains and Advanced.

IIT JEE Syllabus 2026 Overview

JEE Syllabus 2026 for Mains as well as for Advanced exams includes three main subjects: Physics, Chemistry, and Mathematics. While both the JEE Main and JEE Advanced syllabuses are based on Class 11 and 12 NCERT concepts, they differ from each other in terms of depth and difficulty level.

Whereas the JEE Main syllabus aims to test your basic understanding of foundational concepts, the JEE syllabus 2026 for Advanced covers tough topics and trickier problems to test your analytical thinking, problem-solving ability, and how you apply concepts in different questions.

Knowing the difference between the JEE syllabus 2026 for Mains and Advanced helps you study smart, focus on the right topics, and train your mind for both easy and tough questions. Let's now first go through the JEE Main Syllabus 2026.

JEE Main Syllabus

JEE Main is the first stage of the Joint Entrance Examination, conducted by the National Testing Agency (NTA) for admission to engineering colleges like NITs, IIITs, and CFTIs. All students who have passed or are in Class 12 can take this exam, which makes the competition very high. Here, knowing the JEE Main syllabus helps you focus on the topics that are important to score well.

JEE Main syllabus 2026 includes important topics from Physics, Chemistry, and Mathematics of NCERT Classes 11 and 12. The official JEE syllabus for mains is expected to be released by NTA in November 2025. Until then, you can prepare by referring to the previous year's syllabus because the JEE Main Syllabus 2026 is expected to remain the same.

Find the detailed JEE Syllabus 2026 for Mains covering Physics, Maths, and Chemistry here:

Physics JEE Syllabus 2026 For Mains

Physics JEE Syllabus for Mains covers all the basic and important concepts you need to study from Class 11 and 12. It includes topics like Mechanics, Laws of Motion, Work and Energy, Gravitation, Rotations, and Thermodynamics.

Understanding these topics from the Physics JEE syllabus deeply is important because they lay the foundation of this section and help you score good marks in JEE Main. Check the detailed Physics JEE Syllabus for Mains here:

Physics JEE Syllabus For Mains

Units	Topics
Units and Measurements	Units of measurement, System of units, SI Units, fundamental and derived units, least count, significant figures, Errors in measurements. Dimensions of Physics quantities, dimensional analysis and its applications.
Kinematics	The frame of reference, motion in a straight line, speed and velocity, uniform and non-uniform motion, average speed and instantaneous velocity, uniformly accelerated motion, velocity-time, position-time graph, relations for uniformly accelerated motion, relative velocity. Motion in a plane, projectile motion, uniform circular motion



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	applications. vehicle on a level circular road, vehicle on a banked road.
Work, Energy and Power	Work done by a constant force and a variable force, kinetic and potential energies, work-energy theorem, power. The potential energy of a spring, conservation of mechanical energy, conservative and non-conservative forces, motion in a vertical circle. Elastic and inelastic collisions in one and two dimensions.
Rotational Motion	Centre of mass of a two-particle system, centre of mass of a rigid body. Basic concepts of rotational motion, moment of a force, torque, angular momentum, conservation of angular momentum and its applications. The moment of inertia, the radius of gyration, values of moments of inertia for simple geometrical objects, parallel and perpendicular axes theorems and their applications. Equilibrium of rigid bodies, rigid body rotation and equations of rotational motion, comparison of linear and rotational motions.

Gravitation	The universal law of gravitation. Acceleration due to gravity and its variation with altitude and depth. Kepler's law of planetary motion. Gravitational potential energy, gravitational potential. Escape velocity, motion of a satellite, orbital velocity, time period and energy of satellite
Properties of Solids and Liquids	Elastic behaviour, stress-strain relationship, Hooke's Law, Young's modulus, bulk modulus and modulus of rigidity. Pressure due to a fluid column, Pascal's law and its applications, effect of gravity on fluid pressure, viscosity, Stoke's law, terminal velocity, streamline and turbulent flow, critical velocity, Bernoulli's principle and its applications. Surface energy and surface tension, angle of contact, excess of pressure across a curved surface, application of surface tension: drops, bubbles and capillary rise. Heat, temperature, thermal expansion, specific heat capacity, calorimetry, change of state, latent heat. Heat transfer: conduction, convection and radiation.
Thermodynamics	Thermal equilibrium and the concept of temperature, zeroth law of thermodynamics, heat, work and internal energy. The first law of thermodynamics, isothermal and adiabatic processes. The second law of thermodynamics: reversible and irreversible processes.
Kinetic Theory of Gases	Equation of state of a perfect gas, work done on compressing a gas, kinetic theory of gases: assumptions, the concept of pressure, kinetic interpretation of temperature, RMS speed of gas molecules, degrees of freedom, law of equipartition of energy and applications to specific heat capacities of gases, mean free path, Avogadro's number.
Oscillations and Waves	Oscillations and periodic motion: time period, frequency, displacement as a function of time, periodic functions. Simple harmonic motion (S.H.M.) and its equation, phase, oscillations of a spring: restoring force and force constant, energy in S.H.M.: kinetic and potential energies, simple pendulum: derivation of expression for its time period. Wave motion, longitudinal and transverse waves, speed of the travelling wave, displacement relation for a progressive wave, principle of superposition of waves, reflection of waves, standing waves in strings and organ pipes, fundamental mode and harmonics, beats

Electrostatics	Electric charges: conservation of charge, Coulomb's law forces between two point charges, forces between multiple charges, superposition principle and continuous charge distribution. Electric field: electric field due to a point charge, electric field lines, electric dipole, electric field due to a dipole, torque on a dipole in a uniform electric field. Electric flux, Gauss's law and its applications to find fields due to infinitely long uniformly charged straight wire, uniformly charged infinite plane sheet and uniformly charged thin spherical shell. Electric potential and its calculation for a point charge, electric dipole and system of charges, potential difference, equipotential surfaces, electrical potential energy of a system of two point charges and of electric dipole in an electrostatic field. Conductors and insulators, dielectrics and electric polarization, capacitors and capacitance, the combination of capacitors in series and parallel and capacitance of a parallel plate capacitor with and without dielectric medium between the plates, energy stored in a capacitor.
Current Electricity	Electric current: drift velocity, mobility and their relation with electric current, Ohm's law, electrical resistance, I-V characteristics of Ohmic and non-ohmic conductors, electrical energy and power, electrical resistivity and conductivity, series and parallel combinations of resistors, temperature dependence of resistance. Internal resistance, potential difference and emf of a cell, a combination of cells in series and parallel. Kirchhoff's laws and their applications, Wheatstone bridge, Metre Bridge
Magnetic Effects of Current and Magnetism	Biot - Savart law and its application to the current carrying circular loop, Ampere's law and its applications to infinitely long current carrying straight wire and solenoid. Force on a moving charge in uniform magnetic and electric fields, force on a current-carrying conductor in a uniform magnetic field, the force between two parallel currents carrying conductors-definition of ampere, torque experienced by a current loop in a uniform magnetic field: Moving coil galvanometer, its sensitivity and conversion to ammeter and voltmeter. Current loop as a magnetic dipole and its magnetic dipole moment, bar magnet as an equivalent solenoid, magnetic field lines, magnetic field due to a magnetic dipole (bar magnet) along its axis and perpendicular to its axis, torque on a magnetic dipole in a uniform magnetic field, para-, dia- and ferromagnetic substances with examples, the effect of temperature on magnetic properties

Electromagnetic Induction and Alternating Currents	Electromagnetic induction: Faraday's law, induced emf and current, Lenz's law, eddy currents, self and mutual inductance. Alternating currents, peak and RMS value of alternating current/voltage, reactance and impedance, LCR series circuit, resonance, power in AC circuits, wattless current, AC generator and transformer.
Electromagnetic Waves	Displacement current, electromagnetic waves and their characteristics, transverse nature of electromagnetic waves, electromagnetic spectrum (radiowaves, microwaves, infrared, visible, ultraviolet, X-rays, Gamma rays), applications of electromagnetic waves.
Optics	Reflection of light, spherical mirrors, mirror formula. Refraction of light at plane and spherical surfaces, thin lens formula and lens maker formula, total internal reflection and its applications, magnification, power of a lens, combination of thin lenses in contact, refraction of light through a prism, microscope and astronomical telescope (reflecting and refracting) and their magnifying powers. Wave optics: wavefront and Huygens 'Principle, laws of reflection and refraction using Huygens principle. Interference: Young's double-slit experiment and expression for fringe width, coherent sources and sustained interference of light. Diffraction due to a single slit, width of central maximum. Polarization: plane-polarized light, Brewster's law, uses of plane- polarized light and Polaroid.
Dual Nature of Matter and Radiation	Dual nature of radiation, Photoelectric effect, Hertz and Lenard's observations, Einstein's photoelectric equation, particle nature of light. Matter waves: wave nature of particle, de- Broglie relation.
Atoms and Nuclei	Alpha-particle scattering experiment, Rutherford's model of atom, Bohr model, energy levels, hydrogen spectrum. Composition and size of nucleus, atomic masses, mass-energy relation, mass defect, binding energy per nucleon and its variation with mass number, nuclear fission and fusion.
Electronic Devices	Semiconductors, semiconductor diode: I-V characteristics in forward and reverse bias, diode as a rectifier; I-V characteristics of LED, the

photodiode, solar cell, Zener diode, Zener diode as a voltage regulator. Logic gates (OR. AND. NOT. NAND and NOR).

Practicing with **8 Years JEE Physics PYQ** helps students master important concepts, understand exam patterns, and improve problem-solving speed and accuracy in Physics for JEE preparation.

Chemistry JEE Syllabus 2026 For Mains

Chemistry JEE Syllabus 2026 for Mains is divided into three parts: Physical, Inorganic, and Organic Chemistry. It includes important topics like Hydrogen, s-Block Elements, Isolation of Metals, Alcohols, Phenols, Ethers, Polymers, and Environmental Chemistry. Covering these topics well is important for attempting the Chemistry section with accuracy to score well.

Check the detailed Chemistry JEE Syllabus for Mains covering Physical, Inorganic, and Organic Chemistry:

Chemistry JEE Syllabus For Mains	
JEE Main Syllabus for Physical Chemistry (Section A)	
Units	Topics
SOME BASIC CONCEPTS IN CHEMISTRY	Matter and its nature, Dalton's atomic theory, Concept of atom, molecule, element, and compound, Laws of chemical combination, Atomic and molecular masses, mole concept, molar mass, percentage composition, empirical and molecular formulae, Chemical equations, and stoichiometry.
ATOMIC STRUCTURE	Nature of electromagnetic radiation, photoelectric effect, spectrum of the hydrogen atom, Bohr model of a hydrogen atom - its postulates, derivation of the relations for the energy of the electron and radii of the different orbits, limitations of Bohr's model, dual nature of matter, de Broglie's relationship, Heisenberg uncertainty principle, elementary ideas of quantum mechanics, the quantum mechanical model of the atom and its important features, concept of atomic orbitals as one-electron wave functions, variation of ψ and ψ^2 with r for 1s and 2s orbitals, various quantum numbers (principal, angular momentum and magnetic quantum numbers) and their significance, shapes of s, p and d - orbitals, electron spin

	and spin quantum number, rules for filling electrons in orbitals – Aufbau principle, Pauli's exclusion principle and Hund's rule, electronic configuration of elements and extra stability of half-filled and completely filled orbitals.
CHEMICAL BONDING AND MOLECULAR STRUCTURE	Kossel-Lewis approach to chemical bond formation, the concept of ionic and covalent bonds. Ionic Bonding: Formation of ionic bonds, factors affecting the formation of ionic bonds; calculation of lattice enthalpy. Covalent Bonding: Concept of electronegativity, Fajan's rule, dipole moment, Valence Shell Electron Pair Repulsion (VSEPR) theory, and shapes of simple molecules. Quantum mechanical approach to covalent bonding: Valence bond theory - its important features, the concept of hybridization involving s, p, and d orbitals, and resonance. Molecular Orbital Theory - Its important features, LCAOs, types of molecular orbitals (bonding, antibonding), sigma and pi-bonds, molecular orbital electronic configurations of homonuclear diatomic molecules, the concept of bond order, bond length and bond energy. Elementary idea of metallic bonding, hydrogen bonding, and its applications.
CHEMICAL THERMODYNAMICS	Fundamentals of thermodynamics: System and surroundings, extensive and intensive properties, state functions, entropy, types of processes. The first law of thermodynamics - Concept of work, heat, internal energy and enthalpy, heat capacity, molar heat capacity, Hess's law of constant heat summation, Enthalpies of bond dissociation, combustion, formation, atomization, sublimation, phase transition, hydration, ionization, and solution. The second law of thermodynamics - Spontaneity of processes, ΔS of the universe and ΔG of the system as criteria for spontaneity. ΔG° (Standard Gibbs energy change) and equilibrium constant.
SOLUTIONS	Different methods for expressing the concentration of solution - molality, molarity, mole fraction, percentage (by volume and mass both), the vapour pressure of solutions and Raoult's Law- Ideal and nonideal solutions, vapour pressure - composition, plots for ideal and non-ideal solutions, Colligative properties of dilute solutions - a relative lowering of vapour pressure, depression of freezing point, the elevation of boiling point and osmotic pressure, determination of molecular mass using colligative properties, abnormal value of molar mass, van't Hoff factor and its significance

EQUILIBRIUM	Meaning of equilibrium is the concept of dynamic equilibrium. Equilibria involving physical processes: Solid-liquid, liquid-gas, gas-gas and solid-gas equilibria, Henry's law. General characteristics of equilibrium involving physical processes. Equilibrium involving chemical processes: Law of chemical equilibrium, equilibrium constants (K_p and K_c) and their significance, the significance of ΔG and ΔG° in chemical equilibrium, factors affecting equilibrium concentration, pressure, temperature, the effect of catalyst, Le Chatelier's principle. Ionic equilibrium: Weak and strong electrolytes, ionization of electrolytes, various concepts of acids and bases (Arrhenius, Bronsted - Lowry and Lewis) and their ionization, acid-base equilibria (including multistage ionization) and ionization constants, ionization of water, pH scale, common effect, hydrolysis of salts and pH of their solutions, the solubility of sparingly soluble salts, solubility products and buffer solutions.
REDOX REACTIONS AND ELECTROCHEMISTRY	Electronic concepts of oxidation and reduction, redox reactions, oxidation number, rules for assigning oxidation number and balancing of redox reactions. Electrolytic and metallic conduction, conductance in electrolytic solutions, molar conductivities and their variation with concentration, Kohlrausch's law and its applications. Electrochemical cells - Electrolytic and Galvanic cells, different types of electrodes, electrode potentials including standard electrode potential, half-cell and cell reactions, emf of a Galvanic cell and its measurement, Nernst equation and its applications, relationship between cell potential and Gibbs' energy change, dry cell and lead accumulator, fuel cells.
CHEMICAL KINETICS	Rate of a chemical reaction, factors affecting the rate of reactions: concentration, temperature, pressure and catalyst, elementary and complex reactions, order and molecularity of reactions, rate law, rate constant and its units, differential and integral forms of zero and first-order reactions, their characteristics and half-lives, the effect of temperature on the rate of reactions, Arrhenius theory, activation energy and its calculation, collision theory of bi-molecular gaseous reactions (no derivation).
JEE Main Syllabus for Inorganic Chemistry	

Units	Topics
CLASSIFICATION OF ELEMENTS AND PERIODICITY IN PROPERTIES	Modern periodic law and present form of the periodic table, s, p, d, and f block elements, periodic trends in properties of elements of atomic and ionic radii, ionization enthalpy, electron gain enthalpy, valence, oxidation states, and chemical reactivity.
p- BLOCK ELEMENTS	Group -13 to Group 18 Elements General Introduction: Electronic configuration and general trends in physical and chemical properties of elements across the periods and down the groups, unique behaviour of the first element in each group.
d - and f- BLOCK ELEMENTS	Transition Elements - General introduction, electronic configuration, occurrence and characteristics, general trends in properties of the first-row transition elements - physical properties, ionization enthalpy, oxidation states, atomic radii, colour, catalytic behaviour, magnetic properties, complex formation, interstitial compounds, alloy formation, preparation, properties and uses of $K_2Cr_2O_7$ and $KMnO_4$. Inner Transition Elements Lanthanoids - Electronic configuration, oxidation states, and Lanthanoid contraction. Actinoids - Electronic configuration and oxidation states.
COORDINATION COMPOUNDS	Introduction to coordination compounds. Werner's theory, ligands, coordination number, denticity, chelation, IUPAC nomenclature of mononuclear co-ordination compounds, isomerism, Bonding: Valence bond approach and basic ideas of Crystal field theory, colour and magnetic properties, importance of coordination compounds (in qualitative analysis, extraction of metals and in biological systems).
JEE Main Syllabus for Organic Chemistry	
Units	Topics

PURIFICATION AND CHARACTERISATION OF ORGANIC COMPOUNDS	Purification - Crystallization, sublimation, distillation, differential extraction, and chromatography-principles and their applications. Qualitative analysis - Detection of nitrogen, sulphur, phosphorus and halogens. Quantitative analysis (basic principles only) - Estimation of carbon, hydrogen, nitrogen, halogens, sulphur and phosphorus. Calculations of empirical formulae and molecular formulae, numerical problems in organic quantitative analysis,
SOME BASIC PRINCIPLES OF ORGANIC CHEMISTRY	Tetravalency of carbon, shapes of simple molecules - hybridization (s and p): classification of organic compounds based on functional groups and those containing halogens, oxygen, nitrogen and sulphur, homologous series: Isomerism - structural and stereoisomerism. Nomenclature (Trivial and IUPAC) Covalent bond fission - Homolytic and heterolytic, free radicals, carbocations and carbanions, stability of carbocations and free radicals, electrophiles and nucleophiles. Electronic displacement in a covalent bond - Inductive effect, electromeric effect, resonance and hyperconjugation. Common types of organic reactions- Substitution, addition, elimination and rearrangement.
HYDROCARBONS	Classification, isomerism, IUPAC nomenclature, general methods of preparation, properties, and reactions. Alkanes - Conformations: Sawhorse and Newman projections (of ethane), mechanism of halogenation of alkanes. Alkenes - Geometrical isomerism, mechanism of electrophilic addition, addition of hydrogen, halogens, water, hydrogen halides (Markownikoffs and peroxide effect), Ozonolysis, and polymerization. Alkynes - Acidic character, addition of hydrogen, halogens, water, and hydrogen halides, polymerization. Aromatic hydrocarbons - Nomenclature, benzene - structure and aromaticity, mechanism of electrophilic substitution, halogenation, nitration. Friedel-Craft's alkylation and acylation, the directive influence of the functional group in mono-substituted benzene.
ORGANIC COMPOUNDS CONTAINING HALOGENS	General methods of preparation, properties, and reactions, nature of C-X bond, mechanisms of substitution reactions. Uses, environmental effects of chloroform, iodoform, freons, and DDT.

ORGANIC COMPOUNDS CONTAINING OXYGEN	General methods of preparation, properties, reactions, and uses. ALCOHOLS, PHENOLS, AND ETHERS Alcohols: Identification of primary, secondary, and tertiary alcohols, mechanism of dehydration. Phenols: Acidic nature, electrophilic substitution reactions, halogenation, nitration, and sulphonation, Reimer-Tiemann reaction. Ethers: Structure. Aldehyde and Ketones: Nature of carbonyl group, nucleophilic addition to $>C=O$ group, relative reactivities of aldehydes and ketones, important reactions such as - Nucleophilic addition reactions (addition of HCN, NH_3 and its derivatives), Grignard reagent, oxidation, reduction (Wolf Kishner and Clemmensen), the acidity of α-hydrogen. Aldol condensation, Cannizzaro reaction, Haloform reaction, chemical tests to distinguish between aldehydes and ketones. Carboxylic Acids: Acidic strength and factors affecting it
ORGANIC COMPOUNDS CONTAINING NITROGEN	General methods of preparation, properties, reactions and uses. Amines: Nomenclature, classification, structure, basic character, and identification of primary, secondary and tertiary amines and their basic character. Diazonium Salts: Importance in Synthetic Organic Chemistry. UNIT 19: BIOMOLECULES General introduction and importance of biomolecules. CARBOHYDRATES – Classification, aldoses and ketoses, monosaccharides (glucose and fructose), and constituent monosaccharides of oligosaccharides (sucrose, lactose and maltose). PROTEINS - Elementary idea of α-amino acids, peptide bond, polypeptides, proteins: primary, secondary, tertiary and quaternary structure (qualitative idea only), denaturation of proteins, enzymes. VITAMINS – Classification and functions. NUCLEIC ACIDS – Chemical constitution of DNA and RNA, biological functions of nucleic acids. Hormones (General introduction)
PRINCIPLES RELATED TO PRACTICAL CHEMISTRY	Detection of extra elements (Nitrogen, sulphur, halogens) in organic compounds, detection of the following functional groups, hydroxyl (alcoholic and phenolic), carbonyl (aldehyde and ketones), carboxyl and amino groups in organic compounds. The chemistry involved in the preparation of the following: Inorganic compounds, Mohr's salt, potash alum. Organic compounds: Acetanilide, p-nitro acetanilide, aniline yellow, iodoform. The chemistry involved in the titrimetric exercises – acids, bases and the use of indicators, oxalic acid vs $KMnO_4$, Mohr's salt vs $KMnO_4$. Chemical principles involved in the qualitative salt analysis: Cations – Pb^{2+}, Cu^{2+}, Al^{3+}, Fe^{3+}, Zn^{2+}, Ni^{2+}, Ca^{2+}, Ba^{2+}, Mg^{2+}, NH_4^+

Anions – CO_3^{2-} , S^{2-} , SO_4^{2-} , NO_3^- , NO_2^- , Cl^- , Br^- , I^- (Insoluble salts excluded). Chemical principles involved in the following experiments: 1. Enthalpy of solution of CuSO_4 2. Enthalpy of neutralization of strong acid and strong base. 3. Preparation of lyophilic and lyophobic sols. 4. Kinetic study of the reaction of iodide ions with hydrogen peroxide at room temperature.

Practicing with **8 Years JEE Mains Chemistry PYQ** helps students strengthen core concepts, understand important reaction patterns, and improve accuracy by solving real exam-level questions from previous JEE exams.

JEE Maths Syllabus for Mains

JEE Maths Syllabus for Mains covers important topics from both Class 11 and 12. Some key topics are Sets, Relations and Functions, Complex Numbers, Quadratic Equations, Matrices and Determinants. Understanding these topics well helps you solve problems correctly and score good marks in the Mathematics section of JEE Main. Find the detailed unit-wise JEE Maths Syllabus for Mains here:

JEE Maths Syllabus for Mains	
Units	Topics
SETS, RELATIONS AND FUNCTIONS	Sets and their representation; Union, intersection and complement of sets and their algebraic properties; Power set; Relations, type of relations, equivalence relations, functions; one-one, into and onto functions, the composition of functions.
COMPLEX NUMBERS AND QUADRATIC EQUATIONS	Complex numbers as ordered pairs of reals, Representation of complex numbers in the form $a+ib$ and their representation in a plane, Argand diagram, algebra of complex numbers, modulus and argument (or amplitude) of a complex number, Quadratic equations in real and complex number systems and their solutions; Relations between roots and coefficients, nature of roots, the formation of quadratic equations with given roots.

MATRICES AND DETERMINANTS	Matrices, algebra of matrices, type of matrices, determinants and matrices of order two and three, evaluation of determinants, area of triangles using determinants; Adjoint and inverse of a square matrix; Test of consistency and solution of simultaneous linear equations in two or three variables using matrices
PERMUTATIONS AND COMBINATIONS	The fundamental principle of counting, permutations and combinations; Meaning of $P(n, r)$ and $C(n, r)$. Simple applications.
BINOMIAL THEOREM AND ITS SIMPLE APPLICATIONS	Binomial theorem for a positive integral index, general term and middle term and simple applications.
SEQUENCE AND SERIES	Arithmetic and Geometric progressions, insertion of arithmetic, geometric means between two given numbers, Relation between A.M and G.M
LIMIT, CONTINUITY AND DIFFERENTIABILITY	Real-valued functions, algebra of functions; polynomial, rational, trigonometric, logarithmic and exponential functions; inverse functions. Graphs of simple functions. Limits, continuity and differentiability. Differentiation of the sum, difference, product and quotient of two functions. Differentiation of trigonometric, inverse trigonometric, logarithmic, exponential, composite and implicit functions; derivatives of order upto two, Applications of derivatives: Rate of change of quantities, monotonic-Increasing and decreasing functions, Maxima and minima of functions of one variable.
INTEGRAL CALCULAS	Integral as an anti-derivative, Fundamental integrals involving algebraic, trigonometric, exponential and logarithmic functions. Integration by substitution, by parts and by partial fractions. Integration using trigonometric identities. Evaluation of simple integrals of the type $\int dx / (x^2 + a^2),$ $\int dx / (x^2 \pm a^2),$

	$\int dx / (a^2 - x^2),$ $\int dx / \sqrt{(a^2 - x^2)},$ $\int dx / (ax^2 + bx + c),$ $\int dx / \sqrt{(ax^2 + bx + c)},$ $\int (px + q) dx / (ax^2 + bx + c),$ $\int (px + q) dx / \sqrt{(ax^2 + bx + c)},$ $\int \sqrt{(a^2 \pm x^2)} dx,$ $\int \sqrt{(x^2 - a^2)} dx$ The fundamental theorem of calculus, properties of definite integrals. Evaluation of definite integrals, determining areas of the regions bounded by simple curves in standard forms.
DIFFERENTIAL EQUATIONS	Ordinary differential equations, their order and degree, the solution of differential equation by the method of separation of variables, solution of a homogeneous and linear differential equation of the type $dy/dx + p(x) \cdot y = q(x)$
CO-ORDINATE GEOMETRY	Cartesian system of rectangular coordinates in a plane, distance formula, sections formula, locus and situation, the slope of a line, parallel and perpendicular lines, intercepts of a line on the co-ordinate axis. Straight line: Various forms of equations of a line, intersection of lines, angles between two lines, conditions for concurrence of three lines, the distance of a point from a line, co-ordinate of the centroid, orthocentre and circumcentre of a triangle. Circle, conic sections: A standard form of equations of a circle, the general form of the equation of a circle, its radius and centre, equation of a circle when the endpoints of a diameter are given, points of intersection of a line and a circle with the centre at the origin and sections of conics, equations of conic sections(parabola, ellipse and hyperbola) in standard forms.

THREE DIMENSIONAL GEOMETRY	Coordinates of a point in space, the distance between two points, section formula, direction ratios and direction cosines and the angle between two intersecting lines. Equation of a line; Skewlines, the shortest distance between them and its equation.
VECTOR ALGEBRA	Vectors and scalars, the addition of vectors, components of a vector in two dimensions and three-dimensional spaces, scalar and vector products.
STATISTICS AND PROBABILITY	Measures of dispersion; calculation of mean, median, mode of grouped and ungrouped data, calculation of standard deviation, variance and mean deviation for grouped and ungrouped data. Probability: Probability of an event, addition and multiplication theorems of probability, Baye's theorem, probability distribution of a random variable.
TRIGONOMETRY	Trigonometrical identities and trigonometrical functions, inverse trigonometrical functions, and their properties.

Once you go through the detailed JEE Main syllabus of Maths, Physics, and Chemistry, it's important to also check the JEE exam pattern for Mains to plan your preparation accordingly.

Practicing with **8 Years JEE Mains Maths PYQ** helps students strengthen core mathematical concepts, understand exam-level question patterns, and improve speed and accuracy by solving real previous year questions from JEE exams.

JEE Exam Pattern For Mains

JEE Exam Pattern for mains is expected to be the same as in previous years. The JEE Mains usually has three papers: Paper 1 for B.E./B.Tech, Paper 2A for B.Arch, and Paper 2B for B.Plan. All of these exams are conducted in an online mode, except for the Drawing section in B.Arch.

As per the JEE exam pattern for Mains Paper 1, there are usually 75 questions carrying a total of 300 marks. Check the detailed JEE Exam Pattern for Mains below.

JEE Exam Pattern For Mains Paper 1 (B.Tech/B.E.)

Particulars	Details
Mode of Examination	Computer-Based Test (CBT)
JEE Main exam duration	For Non-PwD/PwBD Candidates- 03 hours (180 minutes) For PwD /PwBD Candidates (who are eligible for scribe)-04 hours
Language of Examination	English, Hindi, Assamese, Bengali, Gujarati, Kannada, Malayalam, Marathi, Odia, Punjabi, Tamil, Telugu, and Urdu`
Type of Questions	Objective Type - Multiple Choice Questions (MCQs) and Questions for which the answer is a numerical value, with equal weightage to Mathematics, Physics, and Chemistry
Number of sections and subjects	3 (Physics, Chemistry, and Mathematics)
JEE Main Total Questions	75 (25 each from Physics, Chemistry, Maths)
JEE Main Total Marks	300 (100 per subject)
Question Type	Section A- MCQs (Multiple-Choice Questions) Section B-Numerical Value Questions
Questions per Subject	Section A- 20 Section B- 5

JEE Main Marking Scheme	Each question carries 4 marks
JEE Main Negative Marking	There will be negative marking for incorrect answers in Sections A and B.

After clearing JEE Main, you become eligible to appear for JEE Advanced, which is more challenging and tests your understanding in depth. So, planning your preparation effectively around the JEE Advanced syllabus will help you solve tougher problems and perform better in this next stage of the exam. Let's go through the detailed IIT JEE Advanced Syllabus 2026.

IIT JEE Advanced Syllabus 2026

IIT JEE Advanced Syllabus 2026 is made to test your deeper understanding of Physics, Chemistry, and Mathematics. It focuses not just on what you know, but on how well you can apply concepts, solve problems, and think critically. The questions asked in JEE Advanced are generally tricky and need good reasoning and analytical skills.

Knowing the [JEE Advanced Syllabus](#) properly helps you understand which topics need more practice and which concepts are often asked. With a regular study routine based on the JEE Advanced syllabus, you can prepare wisely, focus on important topics, and boost your chances of passing this challenging exam.

For your reference, we've compiled the detailed IIT JEE Advanced Syllabus 2026 here as per the previous year, as it is expected to be the same.

Physics JEE Syllabus For Advanced

Physics JEE Syllabus for Advanced is based on deeper concepts and topics like Mechanics, Thermodynamics, Laws of Motion, Gravitation, Rotations, Oscillations, Waves, Electrostatics, etc. Understanding these topics deeply is important for you because JEE Advanced questions generally mix multiple concepts to test your reasoning and analytical skills.

JEE Advanced syllabus for Physics is designed to help you apply physics concepts in different situations and while solving numerical problems. Preparing according to this syllabus makes sure you are ready to face the challenging questions of JEE Advanced and can manage time and accuracy effectively during the exam. Find the detailed Physics JEE Syllabus For Advanced below.

JEE Advanced Syllabus for Physics

Topics	Subtopics
General	General Units and dimensions, dimensional analysis; least count, significant figures; Methods of measurement and error analysis for physical quantities pertaining to the following experiments: Experiments based on using Vernier calipers and screw gauge (micrometer), Determination of g using simple pendulum, Young's modulus - elasticity of the material Surface tension of water by capillary rise and effect of detergents. Specific heat of a liquid using calorimeter, focal length of a concave mirror and a convex lens using $u-v$ method, Speed of sound using resonance column, Verification of Ohm's law using voltmeter and ammeter, and specific resistance of the material of a wire using meter bridge and post office box.
Mechanics	Kinematics in one and two dimensions (Cartesian coordinates only), projectiles; Uniform circular motion; Relative velocity. Newton's laws of motion; Inertial and uniformly accelerated frames of reference; Static and dynamic friction; Kinetic and potential energy; Work and power; Conservation of linear momentum and mechanical energy. Systems of particles; Centre of mass and its motion; Impulse; Elastic and inelastic collisions. Rigid body, moment of inertia, parallel and perpendicular axes theorems, moment of inertia of uniform bodies with simple geometrical shapes; Angular momentum; Torque; Conservation of angular momentum; Dynamics of rigid bodies with fixed axis of rotation; Rolling without slipping of rings, cylinders and spheres; Equilibrium of rigid bodies; Collision of point masses with rigid bodies. Forced and damped oscillation (in one dimension), resonance. Linear and angular simple harmonic motions. Hooke's law, Young's modulus. Law of gravitation; Gravitational potential and field; Acceleration due to gravity; Kepler's law, Geostationary orbits, Motion of planets and satellites in circular orbits; Escape velocity. Pressure in a fluid; Pascal's law; Buoyancy; Surface energy and surface tension, angle of contact, drops, bubbles and capillary rise. Viscosity (Poiseuille's equation excluded), Modulus of rigidity and bulk modulus in mechanics. Stoke's law; Terminal velocity, Streamline flow, equation of continuity, Bernoulli's theorem and its applications. Wave motion (plane waves only), longitudinal and transverse waves, superposition of waves; Progressive and stationary waves; Vibration of strings and air columns; Resonance; Beats; Speed of sound in gases; Doppler effect (in sound)

Thermal Physics	Thermal expansion of solids, liquids and gases; Calorimetry, latent heat; Heat conduction in one dimension; Elementary concepts of convection and radiation; Newton's law of cooling; Ideal gas laws; Specific heats (C_v and C_p for monoatomic and diatomic gases); Isothermal and adiabatic processes, bulk modulus of gases; Equivalence of heat and work; First law of thermodynamics and its applications (only for ideal gases); Second law of thermodynamics, reversible and irreversible processes, Carnot engine and its efficiency; Blackbody radiation: absorptive and emissive powers; Kirchhoff's law; Wien's displacement law, Stefan's law.
Electricity and Magnetism	Coulomb's law; Electric field and potential; Electrical potential energy of a system of point charges and of electrical dipoles in a uniform electrostatic field; Electric field lines; Flux of electric field; Gauss's law and its application in simple cases, such as, to find field due to infinitely long straight wire, uniformly charged infinite plane sheet and uniformly charged thin spherical shell. Capacitance; Parallel plate capacitor with and without dielectrics; Capacitors in series and parallel; Energy stored in a capacitor. Electric current; Ohm's law; Series and parallel arrangements of resistances and cells; Kirchhoff's laws and simple applications; Heating effect of current. Biot-Savart's law and Ampere's law; Magnetic field near a current-carrying straight wire, along the axis of a circular coil and inside a long straight solenoid; Force on a moving charge and on a current-carrying wire in a uniform magnetic field. Magnetic moment of a current loop; Effect of a uniform magnetic field on a current loop; Moving coil galvanometer, voltmeter, ammeter and their conversions. Electromagnetic induction: Faraday's law, Lenz's law; Self and mutual inductance; RC, LR, LC and LCR(in series) circuits with d.c. and a.c. sources.
Electromagnetic Waves	Electromagnetic waves and their characteristics. Electromagnetic spectrum (radio waves, microwaves, infrared, visible, ultraviolet, x-rays, gamma rays) including elementary facts about their uses.
Optics	Rectilinear propagation of light; Reflection and refraction at plane and spherical surfaces; Total internal reflection; Deviation and dispersion of light by a prism; Thin lenses; Combinations of mirrors and thin lenses; Magnification. Wave nature of light: Huygen's principle, interference limited to Young's double slit experiment. Diffraction due to a single slit. Polarization of light, plane polarized light; Brewster's law, Polaroids.

Modern Physics	Atomic nucleus; α , β and γ radiations; Law of radioactive decay; Decay constant; Half-life and mean life; Binding energy and its calculation; Fission and fusion processes; Energy calculation in these processes. Photoelectric effect; Bohr's theory of hydrogen-like atoms; Characteristic and continuous X-rays, Moseley's law; de Broglie wavelength of matter waves.
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Practicing with **JEE Advanced Physics PYQ 2027** helps students strengthen core concepts, understand question trends, and improve problem-solving speed and accuracy by solving real previous year questions asked in JEE Advanced exams.

Chemistry JEE Syllabus 2026 for Advanced

Chemistry JEE Advanced syllabus is divided into three main branches: Physical, Inorganic, and Organic Chemistry, the same as the JEE Main. The JEE Advanced syllabus is vast and needs in-depth understanding and regular practice.

So, focusing on all three parts of the JEE Advanced syllabus for Chemistry is important because questions may mix up concepts from different topics. Check the topic-wise JEE Advanced Syllabus for Chemistry here:

Chemistry JEE Syllabus for Advanced	
Topics	Subtopics
General Topics	Concept of atoms and molecules; Dalton's atomic theory; Mole concept; Chemical formulae; Balanced chemical equations; Calculations (based on mole concept and stoichiometry) involving common oxidation-reduction, neutralisation, and displacement reactions; Concentration in terms of mole fraction, molarity, molality and normality.
States of Matter: Gases and Liquids	Gas laws and ideal gas equation, absolute scale of temperature; Deviation from ideality, van der Waals equation; Kinetic theory of gases, average, root mean square and most probable velocities and their relation with temperature; Law of partial pressures; Diffusion of gases. Intermolecular interactions: types, distance dependence, and their effect on properties; Liquids: vapour pressure, surface tension, viscosity.

Atomic Structure	Bohr model, spectrum of hydrogen atom; Wave-particle duality, de Broglie hypothesis; Uncertainty principle; Qualitative quantum mechanical picture of hydrogen atom: Energies, quantum numbers, wave function and probability density (plots only), shapes of s, p and d orbitals; Aufbau principle; Pauli's exclusion principle and Hund's rule.
Chemical Bonding and Molecular Structure	Orbital overlap and covalent bond; Hybridisation involving s, p and d orbitals only; Molecular orbital energy diagrams for homonuclear diatomic species (up to Ne_2); Hydrogen bond; Polarity in molecules, dipole moment; VSEPR model and shapes of molecules (linear, angular, triangular, square planar, pyramidal, square pyramidal, trigonal bipyramidal, tetrahedral and octahedral).
Chemical Thermodynamics	Intensive and extensive properties, state functions, First law of thermodynamics; Internal energy, work (pressure-volume only) and heat; Enthalpy, heat capacity, standard state, Hess's law; Enthalpy of reaction, fusion and vapourization, and lattice enthalpy; Second law of thermodynamics; Entropy; Gibbs energy; Criteria of equilibrium and spontaneity.
Chemical and Ionic Equilibrium	Law of mass action; Significance of K_p and K_c in chemical equilibrium; Equilibrium constant (K_p and K_c) and reaction quotient, Le Chatelier's principle (effect of concentration, temperature and pressure); Solubility product and its applications, common ion effect, pH and buffer solutions; Acids and bases (Bronsted and Lewis concepts); Hydrolysis of salts.
Electrochemistry	Electrochemical cells and cell reactions; Standard electrode potentials; Electrochemical work, Nernst equation; Electrochemical series, emf of galvanic cells; Faraday's laws of electrolysis; Electrolytic conductance, specific, equivalent and molar conductivity, Kohlrausch's law; Batteries: Primary and Secondary, fuel cells; Corrosion.

Chemical Kinetics	Rates of chemical reactions; Order and molecularity of reactions; Rate law, rate constant, half-life; Differential and integrated rate expressions for zero and first order reactions; Temperature dependence of rate constant (Arrhenius equation and activation energy); Catalysis: Homogeneous and heterogeneous, activity and selectivity of solid catalysts, enzyme catalysis and its mechanism.
Solid State	Classification of solids, crystalline state, seven crystal systems (cell parameters a , b , c , α , β , γ), close packed structure of solids (cubic and hexagonal), packing in fcc, bcc and hcp lattices; Nearest neighbours, ionic radii and radius ratio, point defects
Solutions	Henry's law; Raoult's law; Ideal solutions; Colligative properties: lowering of vapour pressure, elevation of boiling point, depression of freezing point, and osmotic pressure; van't Hoff factor.
Surface Chemistry	Elementary concepts of adsorption: Physisorption and Chemisorption, Freundlich adsorption isotherm; Colloids: types, methods of preparation and general properties; Elementary ideas of emulsions, surfactants and micelles (only definitions and examples).
Classification of Elements and Periodicity in Properties	Modern periodic law and the present form of periodic table; electronic configuration of elements; periodic trends in atomic radius, ionic radius, ionization enthalpy, electron gain enthalpy, valence, oxidation states, electronegativity, and chemical reactivity.
Hydrogen	Position of hydrogen in periodic table, occurrence, isotopes, preparation, properties and uses of hydrogen; hydrides – ionic, covalent and interstitial; physical and chemical properties of water, heavy water; hydrogen peroxide-preparation, reactions, use and structure; hydrogen as a fuel.
s-Block Elements	Alkali and alkaline earth metals-reactivity towards air, water, dihydrogen, halogens, acids; their reducing nature including solutions in liquid ammonia; uses of these elements; general characteristics of their oxides, hydroxides, halides, salts of oxoacids; anomalous behaviour of lithium and beryllium; preparation, properties, and uses of compounds of sodium (sodium carbonate, sodium chloride, sodium hydroxide, sodium hydrogen

	carbonate) and calcium (calcium oxide, calcium hydroxide, calcium carbonate, calcium sulphate).
p-Block Elements	Oxidation state and trends in chemical reactivity of elements of groups 13-17; anomalous properties of boron, carbon, nitrogen, oxygen, and fluorine with respect to other elements in their respective groups. Group 13: Reactivity towards acids, alkalis, and halogens; preparation, properties, and uses of borax, orthoboric acid, diborane, boron trifluoride, aluminium chloride, and alums; uses of boron and aluminium. Group 14: Reactivity towards water and halogen; allotropes of carbon and uses of carbon; preparation, properties, and uses of carbon monoxide, carbon dioxide, silicon dioxide, silicones, silicates, zeolites. Group 15: Reactivity towards hydrogen, oxygen, and halogen; allotropes of phosphorous; preparation, properties, and uses of dinitrogen, ammonia, nitric acid, phosphine, phosphorus trichloride, phosphorus pentachloride; oxides of nitrogen and oxoacids of phosphorus. Group 16: Reactivity towards hydrogen, oxygen, and halogen; simple oxides; allotropes of sulfur; preparation/manufacture, properties, and uses of dioxygen, ozone, sulfur dioxide, sulfuric acid; oxoacids of sulfur. Group 17: Reactivity towards hydrogen, oxygen, and metals; preparation/manufacture, properties, and uses of chlorine, hydrogen chloride and interhalogen compounds; oxoacids of halogens, bleaching powder. Group 18: Chemical properties and uses; compounds of xenon with fluorine and oxygen.
d-Block Elements	Oxidation states and their stability; standard electrode potentials; interstitial compounds; alloys; catalytic properties; applications; preparation, structure, and reactions of oxoanions of chromium and manganese.
f-Block Elements	Lanthanoid and actinoid contractions; oxidation states; general characteristics.
Coordination Compounds	Werner's theory; Nomenclature, cis-trans and ionization isomerism, hybridization and geometries (linear, tetrahedral, square planar and octahedral) of mononuclear coordination compounds; Bonding [VBT and CFT (octahedral and tetrahedral fields)]; Magnetic properties (spin-only) and colour of 3d-series coordination compounds; Ligands and spectrochemical series; Stability; Importance and applications; Metal carbonyls.

Isolation of Metals	Metal ores and their concentration; extraction of crude metal from concentrated ores: thermodynamic (iron, copper, zinc) and electrochemical (aluminium) principles of metallurgy; cyanide process (silver and gold); refining.
Principles of Qualitative Analysis	Groups I to V (only Ag^+ , Hg_2^{2+} , Cu^{2+} , Pb^{2+} , Fe^{3+} , Cr^{3+} , Al^{3+} , Ca^{2+} , Ba^{2+} , Zn^{2+} , Mn^{2+} and Mg^{2+}); Nitrate, halides (excluding fluoride), carbonate and bicarbonate, sulphate and sulphide.
Environmental Chemistry	Atmospheric pollution; water pollution; soil pollution; industrial waste; strategies to control environmental pollution; green chemistry.
Basic Principles of Organic Chemistry	Hybridisation of carbon; σ and π -bonds; Shapes of simple organic molecules; aromaticity; Structural and geometrical isomerism; Stereoisomers and stereochemical relationship (enantiomers, diastereomers, meso) of compounds containing only up to two asymmetric centres (R,S and E,Z configurations excluded); Determination of empirical and molecular formulae of simple compounds by combustion method only; IUPAC nomenclature of organic molecules (hydrocarbons, including simple cyclic hydrocarbons and their mono-functional and bi-functional derivatives only); Hydrogen bonding effects; Inductive, Resonance and Hyperconjugative effects; Acidity and basicity of organic compounds; Reactive intermediates produced during homolytic and heterolytic bond cleavage; Formation, structure and stability of carbocations, carbanions and free radicals.
Alkanes	Homologous series; Physical properties (melting points, boiling points and density) and effect of branching on them; Conformations of ethane and butane (Newman projections only); Preparation from alkyl halides and aliphatic carboxylic acids; Reactions: combustion, halogenation (including allylic and benzylic halogenation) and oxidation.
Alkenes and Alkynes	Physical properties (boiling points, density and dipole moments); Preparation by elimination reactions; Acid catalysed hydration (excluding the stereochemistry of addition and elimination); Metal acetylides; Reactions of alkenes with KMnO_4 and ozone; Reduction of alkenes and alkynes; Electrophilic addition reactions of alkenes with X_2 , HX , HOX ,

	(X=halogen); Effect of peroxide on addition reactions; cyclic polymerization reaction of alkynes.
Benzene	Structure; Electrophilic substitution reactions: halogenation, nitration, sulphonation, FriedelCrafts alkylation and acylation; Effect of directing groups (monosubstituted benzene) in these reactions.
Phenols	Physical properties; Preparation, Electrophilic substitution reactions of phenol (halogenation, nitration, sulphonation); Reimer-Tiemann reaction, Kolbe reaction; Esterification; Etherification; Aspirin synthesis; Oxidation and reduction reactions of phenol.
Alkyl Halides	Rearrangement reactions of alkyl carbocation; Grignard reactions; Nucleophilic substitution reactions and their stereochemical aspects.
Alcohols	Physical properties; Reactions: esterification, dehydration (formation of alkenes and ethers); Reactions with: sodium, phosphorus halides, $\text{ZnCl}_2/\text{concentrated HCl}$, thionyl chloride; Conversion of alcohols into aldehydes, ketones and carboxylic acids.
Ethers	Preparation by Williamson's synthesis; C-O bond cleavage reactions.
Aldehydes and Ketones	Preparation of: aldehydes and ketones from acid chlorides and nitriles; aldehydes from esters; benzaldehyde from toluene and benzene; Reactions: oxidation, reduction, oxime and hydrazone formation; Aldol condensation, Cannizzaro reaction; Haloform reaction; Nucleophilic addition reaction with RMgX , NaHSO_3 , HCN , alcohol, amine.
Carboxylic Acids	Physical properties; Preparation: from nitriles, Grignard reagents, hydrolysis of esters and amides; Preparation of benzoic acid from alkylbenzenes; Reactions: reduction, halogenation, formation of esters, acid chlorides and amides.
Amines	Preparation from nitro compounds, nitriles and amides; Reactions: Hoffmann bromamide degradation, Gabriel phthalimide synthesis;

	Reaction with nitrous acid, Azo coupling reaction of diazonium salts of aromatic amines; Sandmeyer and related reactions of diazonium salts; Carbylamine reaction, Hinsberg test, Alkylation and acylation reactions.
Haloarenes	Reactions: Fittig, Wurtz-Fittig; Nucleophilic aromatic substitution in haloarenes and substituted haloarenes (excluding benzyne mechanism and cine substitution).
Biomolecules	Carbohydrates: Classification; Mono- and di-saccharides (glucose and sucrose); Oxidation; Reduction; Glycoside formation and hydrolysis of disaccharides (sucrose, maltose, lactose); Anomers. Proteins: Amino acids; Peptide linkage; Structure of peptides (primary and secondary); Types of proteins (fibrous and globular). Nucleic acids: Chemical composition and structure of DNA and RNA.
Polymers	Types of polymerization (addition, condensation); Homo and copolymers; Natural rubber; Cellulose; Nylon; Teflon; Bakelite; PVC; Bio-degradable polymers; Applications of polymers.
Chemistry in Everyday Life	Drug-target interaction; Therapeutic action, and examples (excluding structures), of antacids, antihistamines, tranquilizers, analgesics, antimicrobials, and antifertility drugs; Artificial sweeteners (names only); Soaps, detergents, and cleansing action.
Practical Organic Chemistry	Detection of elements (N, S, halogens); Detection and identification of the following functional groups: hydroxyl (alcoholic and phenolic), carbonyl (aldehyde and ketone), carboxyl, amino and nitro.

Practicing with **JEE Advanced Chemistry PYQ 2027** helps students strengthen core concepts, understand reaction patterns, and improve accuracy by solving real previous year questions asked in JEE Advanced exams.

JEE Maths Syllabus for Advanced

Same as JEE Main, the Mathematics section of JEE Advanced covers both Class 11 and Class 12 topics, but by focusing more on problem-solving and higher-order thinking. JEE Maths Syllabus for Advanced aims to test your understanding, reasoning, and analytical skills.

So when you know all the topics, it becomes easy for you to study, revise, and solve tricky questions in the exam. Check out the detailed JEE Maths Syllabus for Advanced here:

JEE Maths Syllabus for Advanced	
Topics	Subtopics
Sets, Relations and Functions	Sets and their representations, different kinds of sets (empty, finite and infinite), algebra of sets, intersection, complement, difference and symmetric difference of sets and their algebraic properties, De-Morgan's laws on union, intersection, difference (for finite number of sets) and practical problems based on them. Cartesian product of finite sets, ordered pair, relations, domain and codomain of relations, equivalence relation Function as a special case of relation, functions as mappings, domain, codomain, range of functions, invertible functions, even and odd functions, into, onto and one-to-one functions, special functions (polynomial, trigonometric, exponential, logarithmic, power, absolute value, greatest integer etc.), sum, difference, product and composition of functions.
Algebra	Algebra of complex numbers, addition, multiplication, conjugation, polar representation, properties of modulus and principal argument, triangle inequality, cube roots of unity, geometric interpretations. Statement of fundamental theorem of algebra, Quadratic equations with real coefficients, relations between roots and coefficients, formation of quadratic equations with given roots, symmetric functions of roots. Arithmetic and geometric progressions, arithmetic and geometric means, sums of finite arithmetic and geometric progressions, infinite geometric series, sum of the first n natural numbers, sums of squares and cubes of the first n natural numbers. Logarithms and their properties, permutations and combinations, binomial theorem for a positive integral index, properties of binomial coefficients
Matrices	Matrices as a rectangular array of real numbers, equality of matrices, addition, multiplication by a scalar and product of matrices, transpose of a matrix, elementary row and column transformations, determinant of a square matrix of order up to three, adjoint of a matrix, inverse of a square matrix of order up to three, properties of these matrix operations, diagonal, symmetric and skew-symmetric matrices and their properties, solutions of simultaneous linear equations in two or three variables.

Probability and Statistics	Random experiment, sample space, different types of events (impossible, simple, compound), addition and multiplication rules of probability, conditional probability, independence of events, total probability, Bayes Theorem, computation of probability of events using permutations and combinations. Measure of central tendency and dispersion, mean, median, mode, mean deviation, standard deviation and variance of grouped and ungrouped data, analysis of the frequency distribution with same mean but different variance, random variable, mean and variance of the random variable.
Trigonometry	Trigonometric functions, their periodicity and graphs, addition and subtraction formulae, formulae involving multiple and sub-multiple angles, general solution of trigonometric equations. Inverse trigonometric functions (principal value only) and their elementary properties.
Analytical Geometry	Two dimensions: Cartesian coordinates, distance between two points, section formulae, shift of origin. Equation of a straight line in various forms, angle between two lines, distance of a point from a line; Lines through the point of intersection of two given lines, equation of the bisector of the angle between two lines, concurrency of lines; Centroid, orthocentre, incentre and circumcentre of a triangle. Equation of a circle in various forms, equations of tangent, normal and chord. Parametric equations of a circle, intersection of a circle with a straight line or a circle, equation of a circle through the points of intersection of two circles and those of a circle and a straight line. Equations of a parabola, ellipse and hyperbola in standard form, their foci, directrices and eccentricity, parametric equations, equations of tangent and normal. Locus problems.
	Three dimensions: Distance between two points, direction cosines and direction ratios, equation of a straight line in space, skew lines, shortest distance between two lines, equation of a plane, distance of a point from a plane, angle between two lines, angle between two planes, angle between a line and the plane, coplanar lines.
Differential Calculus	Limit of a function at a real number, continuity of a function, limit and continuity of the sum, difference, product and quotient of two functions, L'Hospital rule of evaluation of limits of functions. Continuity of composite functions, intermediate value property of continuous functions. Derivative of a function, derivative of the sum, difference, product and quotient of two functions, chain rule, derivatives of polynomial, rational, trigonometric, inverse trigonometric,

	exponential and logarithmic functions. Tangents and normals, increasing and decreasing functions, derivatives of order two, maximum and minimum values of a function, Rolle's theorem and Lagrange's mean value theorem, geometric interpretation of the two theorems, derivatives up to order two of implicit functions, geometric interpretation of derivatives
Integral Calculus	Integration as the inverse process of differentiation, indefinite integrals of standard functions, definite integrals as the limit of sums, definite integral and their properties, fundamental theorem of integral calculus. Integration by parts, integration by the methods of substitution and partial fractions, application of definite integrals to the determination of areas bounded by simple curves. Formation of ordinary differential equations, solution of homogeneous differential equations of first order and first degree, separation of variables method, linear first order differential equations.
Vectors	Addition of vectors, scalar multiplication, dot and cross products, scalar and vector triple products, and their geometrical interpretations.

After understanding the complete JEE exam syllabus, check the JEE exam pattern for advanced. It helps you to get familiar with the number of questions, the marking scheme, and the types of questions you will see in the Advanced paper.

Practicing with **JEE Advanced Maths PYQ 2027** helps students strengthen core mathematical concepts, improve problem-solving speed, and understand real exam patterns by solving previous year questions asked in JEE Advanced exams.

JEE Exam Pattern for Advanced

JEE Advanced 2026 is a computer-based test (CBT) expected to be held by the IIT Roorkee in May. The exam has two papers, Paper 1 and Paper 2, each of three hours duration (four hours for PwD candidates). Both papers will have questions based on the JEE exam syllabus for Advanced, with around 18 questions from each subject, making a total of 54 questions per paper. The questions will include multiple-choice questions (single or multiple correct answers), numerical value questions, and matching-type questions.

As per the JEE exam pattern for Advanced, the marking system will include full, partial, and zero marks based on the accuracy of your answers. To understand better, check the detailed JEE Advanced Exam Pattern 2026 here:

JEE Exam Pattern for Advanced

Particulars	Details
Mode of Examination	Computer-Based Test (CBT)
Number of Papers	Two (Paper 1 and Paper 2)
Duration of Each Paper	3 hours (4 hours for PwD candidates)
Subjects Covered	Physics, Chemistry, Mathematics
Number of Questions	54 per paper (18 questions per subject)
Types of Questions	MCQs (single/multiple answers), Numerical Value, Matching-Type
Marking Scheme	Full, Partial, and Zero marks based on correctness
Medium of Examination	English and Hindi

How to Cover the JEE Syllabus 2026 for Mains and Advanced Together

Studying for JEE Main and Advanced together can be a game-changer for you, but at the same time, it can feel tough. But when you have a structured timetable and a reliable set of [JEE Modules Books](#), this becomes easier.

To do this, it is important to focus on common topics between both JEE syllabuses, strengthen your basics, and practice regularly. To understand the JEE exam syllabus for Mains and Advanced, you can solve [JEE Previous Year Papers](#) because they cover questions from all the key chapters. Along with this, here are some tips that can help you cover the JEE Syllabus 2026 for Mains and Advanced together:

- **Start with Common Topics:** Look at the JEE exam syllabus for Main and Advanced and note the topics that appear in both first. This helps you cover common topics and will also save time.

- **Make a Balanced Study Plan:** Divide your day between Physics, Chemistry, and Maths. Solve basic questions from the JEE Main syllabus. After that, try to solve high-level questions from the JEE Advanced syllabus. You can use [JEE Question Banks](#) by PW to find plenty of questions to practice.
- **Use NCERT and Reference Books:** Always start studying with NCERT books, especially for Chemistry, to understand the basic concepts deeply. After that, move to other reference books for tougher problems.
- **Practice Regularly:** Solve JEE previous year papers of both exams to identify important chapters and repeated questions.
- **Revise Smartly:** Make short notes and formula sheets for both JEE Main and Advanced separately to revise faster. Daily revision makes sure that you don't forget what you studied. You can also use [JEE Revision Books](#) by PW to make your revision sessions more exam-focused.
- **Take Mock Tests:** Taking the JEE Test Series helps you practice full-length tests under exam-like time limits. After solving question papers, analyze your mistakes and improve them.

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is a complete practice resource for JEE 2027 aspirants. It includes chapterwise and topicwise solved previous year questions with detailed solutions to strengthen concepts and improve problem-solving skills. Covering both JEE Main and JEE Advanced, this combo helps identify exam trends, master important topics, and build confidence through extensive PYQ-based preparation.

JEE Syllabus 2026 and Exam Pattern FAQs

Q.1. Are the JEE exam syllabus for Mains and Advanced the same?

Ans. No. While both exams are based on Class 11 and 12 topics, the JEE Advanced syllabus is more detailed and tougher than Mains, which focuses on basic concepts.

Q.2. What subjects are covered in the IIT JEE Syllabus 2026?

Ans. The IIT JEE syllabus 2026 consists of three main subjects: Physics, Chemistry, and Mathematics.

Q.3. Who will release the JEE syllabus 2026?

Ans. The official JEE Main syllabus will be released by NTA. IIT Roorkee is expected to announce the JEE Advanced syllabus 2026.

Q.4. How can I prepare for the JEE Mains and Advanced syllabus together?

Ans. Start with common topics from both JEE exam syllabuses. Use NCERT books for basics, solve JEE previous year papers, do daily revisions, and attempt mock tests to improve speed and confidence.

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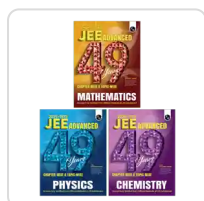
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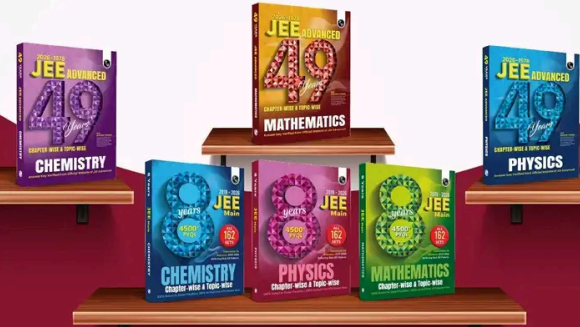
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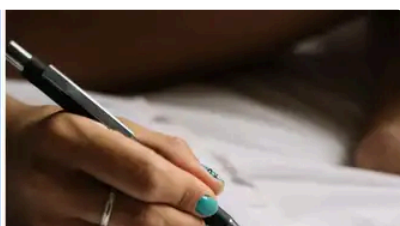
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